

BB 626: Modeling biological systems and processes

Date : 17 Feb 2025

Max. Marks : 5

Simulate a polymer with initial length $l_0 = 100$, rate of polymerization $k_f = 10$ and rate of depolymerization $k_r = 5$ using kinetic Monte-Carlo (Gillespie) algorithm. And do the following quantities:

1. Plot the polymer length, l , vs time for N different polymers (plot in the same graph; not N different graphs).
2. Calculate the ensemble average $\langle l \rangle$ as a function of time and find the slope. Does the slope make sense?
3. Calculate variance and plot $\sigma^2 = \langle l^2 \rangle - \langle l \rangle^2$ as a function of time. What is the slope and does it make sense?

Email the completed assignment to : ranjith.courses@gmail.com before the mid-semester test date (on or before 24 Feb 2025)